

Daniel Dallal

laviedallal@gmail.com | (650) 815-5260 | San José, CA

PROJECTS & ENGINEERING EXPERIENCE

Golden Goose Delivery Aircraft • GooseWorks Senior Capstone Team, SJSU Aug 2026 – Present

- Successfully advocated for twin-boom aircraft configuration, justifying central payload placement, two-engine layout, and aerodynamic flexibility to the 35 lb MTOW, 15 lb payload, and 18 mi range objectives.
- Built a five-tab Excel aircraft sizing model, verifying equations, unit consistency, and realistic input/output relations while adding selectable batteries, mission goals, wing loading, stall speed, tail sizing, lift, drag, power, range, endurance, and climb calculations.
- Used preliminary model results to expose the tradeoff between increased power requirements, battery mass, payload capacity, and allowable airframe weight; preparing the model for validation against test-aircraft data and implementation in MATLAB.

Aerodynamics Intern • Spartan Racing Formula SAE, San José, CA May 2026 – Present

- Coordinated with team lead on ownership of FW development; transferred the existing design into the updated SolidWorks vehicle baseline and independently evaluated 10 CFD iterations using StarCCM+.
- Analyzed pressure and velocity fields to identify a large high-pressure region and flow separation behind the flaps and beneath the mainplane, linking these features to reduced downforce and worsened tire-squirt behavior.
- Created a 3-page CFD workflow guide and 23-slide onboarding presentation used by at least three designers to verify CAD preparation, simulation setup, remote execution, convergence monitoring, and post-processing.

High-Power Rocketry - Level 1 & 2 Certification • AE 166 Rocketry, San José, CA Jan 2026 – May 2026

- Designed, fabricated, and simulated L1 and L2 rockets using SolidWorks and OpenRocket; earned Tripoli L1 and NAR L2 certifications on first attempts.
- Applied L1 stability lessons to design the L2 around a 1.6 - 1.8 caliber margin, accommodate uncertain avionics mass, and preserve motor flexibility through revised fin geometry and a custom 3D-printed avionics tray.
- Replaced the planned J435 motor with a J270 after assessing structural and PETG nose-cone vulnerability in transonic regime; rechecked mass, CG, CP, and stability before achieving 1,295 m measured apogee.

6-DoF Aircraft Spin Dynamics Model • AE 140 Rigid Body Dynamics, SJSU Mar 2026 – May 2026

- Led a four-person project developing a MATLAB 6-DoF aircraft spin model; independently derived the nonlinear equations of motion from $F = ma$, $H = I\omega$, and $M = dH/dt$ and using ode45 integration.
- Debugged team code and validated stabilizing, sustained-spin, and divergent-spin cases using a custom 3D visualization, research findings, time-history plots; received an A.

Sustainable Aviation Technologies Research Paper • Self-Initiated, San José, CA Mar 2025 – Jan 2026

- Reworked a 31-page aviation sustainability paper into a focused 10-page technical paper evaluating sustainable aviation fuels, eVTOL aircraft, and electric propulsion.
- Researched and fact-checked the eVTOL section using government reports, academic research, and aerospace publications; reformatted the paper into a journal-style technical document and received an A.

Autonomous Robot “Edmund” • Eng 10, San José State University, San José, CA Mar 2024 – May 2024

- Designed and iterated the mechanical subsystem of a six-person autonomous robot, balancing arm mass, pressing force, servo torque, sensor protection, and collision behavior across three arm designs.
- Added travel limiters and asymmetric arm geometry to resolve over-travel and wall-collision failures; after the team isolated a faulty command unit, the robot completed all five autonomous tasks in 29 seconds.

Wind Turbine Blade & Structure • San José State University, San José, CA Feb 2024 – Mar 2024

- Designed and built a multi-tier balsa wood and foam-board truss tower under height, mass, and stiffness constraints, using diagonal and cross-bracing to improve stiffness and torsional resistance.
- Measured 14.5 N/mm stiffness at 0.235 kg, supported a 4.1 kg mass in deflection testing, and generated 2.31 W at 700–800 RPM versus a 1.4 W theoretical prediction.

WORK EXPERIENCE

Customer Service Associate II • Team Sheepher, Menlo Park, CA

Jun 2022 – Present

- Redesigned four team-standard Google Sheets systems using data validation, dropdowns, conditional formatting, standardized fields, completion tracking, and accountability fields, improving visibility of incomplete customer-service tasks.
- Standardized English and Spanish customer-email templates and authored comprehensive onboarding documentation covering department workflows and responsibilities.
- Trained 12+ seasonal employees through hands-on shadowing and was promoted to Associate II after four years for sustained process-improvement, auditing, documentation, and training contributions.

LEADERSHIP & COMMUNITY SERVICE

Founding Member & Lead “Wrench” • MA On the Move, Menlo Park, CA

Sep 2018 – May 2022

- Helped establish an official student bicycle-repair and community-service club, serving as one of the experienced Lead Wrenches trusted to diagnose repairs and guide newer members.
- Diagnosed and repaired derailleur, drivetrain, brake and shifter cable, tire, handlebar, and general mechanical failures through hands-on practice and peer instruction.
- Independently refurbished at least 20 donated bicycles and helped the club distribute several dozen repaired bicycles to students in need.

EDUCATION

B.S. Aerospace Engineering • San José State University, San José, CA

Graduating May 2027

Relevant Coursework: Aerodynamics • Flight Mechanics • Rigid Body Dynamics • Fluid Dynamics • Rocketry • Control Systems Design • Statics, Structures & Materials • Aircraft Design • Aerothermodynamics

TECHNICAL SKILLS

Software: MATLAB, Simulink, SolidWorks, STAR-CCM+, OpenRocket, Excel, Sabalcore

Engineering Methods: Preliminary aircraft sizing and performance modeling • CFD setup and post-processing •

Aerodynamic and stability analysis • 6-DoF equations-of-motion derivation • Rigid-body dynamics • State-space modeling • ode45 numerical integration • Data acquisition and post-processing • Composite fabrication • Experimental testing • Structural hand calculations • Stress, buckling, and free-body-diagram analysis

Languages: English (native), Hebrew (fluent), Spanish (fluent)